

Prospective weighted-CM relation search for NIST P-192

2026-10-06

Status and decision

This bundle freezes a new protocol, EXP-SCURVE-1a8daf. **No scientific search, timing run, map census, ECDLP recovery, or replication has been performed.** Every planned run is pending and the current claim ceiling is PROTOCOL_DESIGNED_UNEXECUTED.

The native prerequisite is [crypto PR #1507](#), reviewed here at commit 315c7b417305db4f34ad01cb07c18ca2c55181ee. The prior archive is [crypto-autoresearcher PR #1942](#), reviewed at 5ed7dfbbc02a975e5919bc03b3020f1931e30813. The prior EXP-SCURVE-647ade is a different, immutable ($L \leq 2^{48}$) generator-word census. This protocol does not extend it, rerun it, or consume its run budget.

Labels used below are strict:

- **Verified (protocol):** local YAML parses, links resolve, arithmetic equalities and derived box counts pass the non-scientific checker.
- **Derived:** follows exactly from the frozen integers and equations.
- **Proposed:** an algorithm, budget, threshold, or future measurement.
- **Measured:** none.

Why this is not another blind BFS

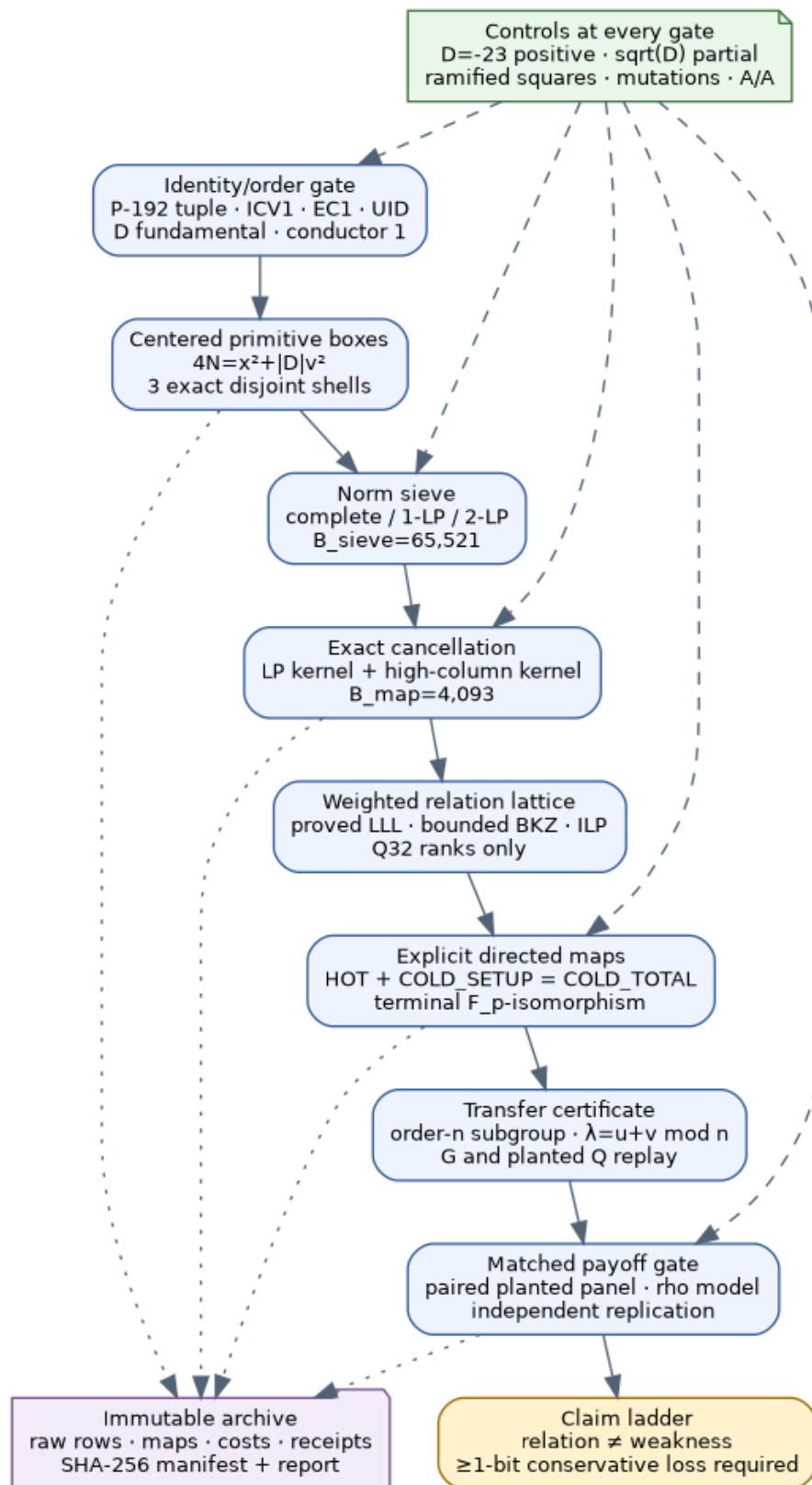
A bounded breadth-first isogeny walk allocates effort by graph distance. It can inventory nearby curves, but it does not preferentially search for a cheap loop returning to the source. This protocol instead enumerates principal quadratic integers, whose ideal factorizations are loops by construction. It then uses a larger algebraic relation base for yield, cancels every prime above the smaller mappable base exactly, and optimizes only certified relation combinations.

That design separates four questions that are easy to conflate:

1. Does an abstract principal ideal relation exist?
2. Can its large-prime and high-degree support be cancelled exactly?
3. Can every remaining directed isogeny and the terminal isomorphism be built?
4. Does the resulting action make the same subgroup DLP cheaper after all search, construction, transfer, and verification costs are charged?

Only the fourth question can support a weakness claim.

P-192 weighted-CM protocol — all scientific stages pending



Exact object and CM assumptions

The curve is the exact short-Weierstrass NIST P-192 / secp192r1 tuple over

[$p=6277101735386680763835789423207666416083908700390324961279$.]

Its coefficient is $(a=p-3)$, its published $(b,G,n,h=1)$ are recorded in the experiment specification, and its trace is

[$t=31607402316713927207482677199, \text{\\quad } n=p+1-t$.]

The three exact identity strings are:

identity	value
ICV1	icv1-fp192-t31607402316713927207482677199-52e4af59
EC1	EC1P192Cp192h5531c4a08bdb
UID	urn:ec-record:1:sha256:5531c4a08bdb64b6e86a6e30e9a08aa57edef7af15ac5f6d4d2a83a53bf2f646

The proposed order gate must independently replay

[$D=t^2-4p=-24109379060336110122544161233113975664949272517896865359515$]

and

[$D=-5 \cdot t \cdot 14140398275856956083603613626459809774163796198179979683$.]

Write $D_{\pi}=t^2-4p$ for the Frobenius-order discriminant, D_K for the fundamental field discriminant, and D_{End} for the geometric endomorphism-order discriminant. The archived derivation says the native certificate proves $D_{\pi}=D_K=D_{\text{End}}=D$: p does not divide t , so the curve is ordinary, and D_{π} is squarefree and fundamental, so $\mathbb{Z}[\pi]$ is already maximal and no larger geometric order can contain it. Thus the claimed result is $\text{End}_{\overline{\mathbb{F}_p}}(E)=\mathbb{Z}[\pi]=0_D$. That chain is an execution prerequisite here, not a conclusion imported from the phrase “CM discriminant” or merely from an archive URL. Since $D<-4$, the unit group must also be checked to be exactly $\{+1, -1\}$.

Centered deterministic search

For $(\alpha=u+v\pi)$, with $(\pi^2-t\pi+p=0)$,

[$N(\alpha)=u^2+uv+pv^2$.]

Set $(x=2u+tv)$. Then

[$4N(\alpha)=x^2+Dv^2, \text{\\quad } x \equiv tv \pmod{2}$.]

For nonscalars, choose $(v>0, x \not\equiv 0)$. Because the only units are (± 1) , this quotients global sign and conjugation; $(x=0)$ is retained as a boundary representative. Primitivity is **exactly** $(\gcd(|u|,v)=1)$. Using $(\gcd(x,v))$ would be wrong: the important $(\sqrt{D}=2\pi-t)$ control has $(x=0, v=2)$ but is primitive because (t) is odd.

The inclusive boxes and exact raw counts are:

box	$(1 \leq v \leq V)$	$(0 \leq x \leq X)$	raw pairs	disjoint shell pairs
BOX-0	8	65,536	262,148	262,148
BOX-1	64	1,048,576	33,554,464	33,292,316
BOX-2	256	4,194,304	536,871,040	503,316,576

Traversal is shell, then increasing (v) , then increasing parity-compatible (x) . Logical shards contain 1,048,576 records, except a final short shard. The specification fixes domain separators, length fields, big-endian integer encoding, shard digests, ordered shell roots, and a family root. Parallel execution cannot alter the logical bytes. This explicitly repairs the digest ambiguity identified in review of the prior archive.

For $(v \neq 0)$, the equation derives

[$N(\alpha) \geq \lceil |D|/4 \rceil = 6027344765084027530636040308278493916237318129474216339879$.]

The earlier (2^{48}) word boundary therefore cannot explore these principal norms. That is the mathematical reason to change representations.

Smoothness, partials, and exact recombination

The proposed algebraic sieve base contains every split or ramified prime ($\ell \leq 521$). The final mappable base contains the analogous primes ($\ell \leq 4093$). This enlargement is part of this new protocol version, not an implementation default inherited from the old ($\ell \leq 113$) census.

For a split prime, the smaller root (r_0) of $(r^2 - tr + p = 0 \bmod \ell)$ names the positive ideal $(\ell, \pi - r_0)$; the other root is its conjugate and has negative signed coordinate. Ramified coordinates are stored modulo two, with rational scalar content recorded separately. Exact ideal valuations, including prime powers, come from ideal arithmetic/HNF—not from rational divisibility alone.

After base division, retain only:

- complete rows with residual 1;
- one-large-prime rows with a proven split prime ($65 \leq q \leq 2^{31} - 1$); or
- two-large-prime rows with exactly $(q_1 q_2)$, including multiplicity, both proven split primes in that interval.

Partials are not relations. Their signed oriented large-prime rows form an exact integer matrix (L). A graph cycle may propose combinations, but only an independently replayed arbitrary-precision solution of $(L^T c = 0)$ is admitted. The same exact-kernel process cancels every algebraic-base column above 4,093. No high-degree edge receives an implicit zero cost.

The relation encoding is byte-exact and binds the curve UID, immutable algebraic factor-base digest, shell-local index, centered point, norm factorization, scalar content, increasing prime coordinates, explicit LP incidence, and arbitrary-precision signed exponents. A separate map-support manifest can evolve without changing algebraic relation IDs. One algebraic ID covers a conjugate pair, while **:positive** and **:negative** directed variants remain separate for map construction and cost.

Both LP cancellation and high-column elimination require saturated integer kernel bases with HNF/SNF certificates. Their kernels are infinite; only the frozen coefficient-bounded search is exhausted. For a combined vector $\mathbf{z}$, the verifier forms the exact oriented ideal $J(\mathbf{z})$, finds a normalized primitive $\mathbf{beta} = \mathbf{a} + \mathbf{b} \cdot \mathbf{pi}$, and proves $\text{HNF}(J) = \text{HNF}(\mathbf{beta})$ and $N(\mathbf{beta}) = \text{product_split}(\ell^{|\mathbf{abs}(\mathbf{z}_{\ell})|}) \cdot \text{product_ram}(\ell^{|\mathbf{r}_{\ell}|})$, where each ramified $\mathbf{r}_{\ell}$ is the projected bit in $\{0, 1\}$, not its pre-projection signed valuation. It does not assume a formal quotient of source generators is already the primitive generator.

Weighted solvers and costs

The exact row lattice is explored by:

- proved LLL with $(\delta = 99/100, \eta = 51/100)$, 256-bit precision, and a 10,000,000-swap cap in forward and reverse row order;
- deterministic single-thread BKZ at block sizes 20, 30, and 40, two tours and 10,000,000 enumeration nodes per tour; and
- deterministic single-thread ILP with $(|c_i| \leq 2^{20})$, 50,000,000 nodes, zero requested gap, and exact replay of every feasible result.

Solver exhaustion is censored. **optimal** is permitted only with an independently checked zero-gap/proof record; otherwise the label is **BEST_FOUND_WITHIN_BUDGET**.

Each realized directed edge and composite reports the raw tuple

[$(\mathrm{FpM}, \mathrm{FpS}, \mathrm{FpI}, \mathrm{FpAddSub}, \mathrm{polyCoeffM}, \mathrm{logicalBytesRW})$.]

HOT is online evaluation and required conversions/checks. **COLD_SETUP** is root/kernel/map/dual construction, verification, serialization, and cleanup from an empty experiment cache. Every record must satisfy componentwise

[$\mathrm{COLD_TOTAL} = \mathrm{COLD_SETUP} + \mathrm{HOT}$.]

The source UID, target UID, root, orientation, kernels, maps, and endpoints key an edge record. A median cost for a degree or an undirected prime is never a measured path cost. Relation discovery and solver cost remain separate and are included in a whole-method cold comparison unless an explicit public precomputation reuse count is also reported.

Q32 weights normalize FpM to $(2^{\{32\}})$, use ties-to-even rounding, and are derived from 16 deterministic cases with one warmup and 11 ABBA repetitions. They are **ranking only**. They cannot alter membership, acceptance, stopping, or a security claim. The fixed calibration panel receives medians, MAD, IQR, range, and leave-one-case-out rank sensitivity—not an iid population interval.

Map, subgroup, and terminal certificates

Every promoted candidate must archive:

1. every source/target curve identity and coefficient tuple;
2. **e11**, both roots, a global orientation proof that $\ker(\phi)$ is the separable cyclic order-**e11** subgroup scheme $E[\mathbf{e11}] \cap \ker(\pi - [r]) = E[(\mathbf{e11}, \pi - r)]$ on the actual source—not the generally much larger full kernel of $\pi - [r]$ —plus kernel polynomial, rational map, exceptional divisor, and dual-composition identity;
3. every intermediate model and the final $(\mathbf{F}_p)$ -isomorphism back to the exact source UID—not merely an equal (j)-invariant;
4. the exact ordered edge circuit—not an infeasible expanded degree-about- 2^{192} rational function—normalized primitive $\mathbf{beta}=(\mathbf{a}, \mathbf{b})$, norm, relation, independent terminal unit signs, and both conjugate eigenvalues;
5. subgroup replay on (G) and every planted $(Q=[d]G)$, including $(\phi(Q)=[d]\phi(G))$, dual identities, and $(\psi(G)=[\lambda]G, \psi(Q)=[\lambda]Q)$.

The transfer cost includes forward maps, applicable duals, conversions, exceptional handling, terminal isomorphism, validation, and return to the source representation. Moving a DLP to an isogenous curve does not make it easier by itself.

The primary consumer is an explicitly implemented automorphism-quotiented rho walk. A certified order of **lambda** requires a factorization/order certificate for **n-1**; order unknown blocks the orbit-payoff lane. The walk must prove equivariance and charge orbit canonicalization, map/inverse evaluation, partitioning, cycle detection, distinguished points, exceptions, and recovery. No ideal square-root-of-action-size gain is assumed. Explicit orbit enumeration is capped at 64; larger actions need a separately certified non-enumerative consumer. A decomposition alternative must certify a balanced basis of $\langle (\mathbf{n}, 0), (-\mathbf{lambda}, 1) \rangle$ and charge the actual multi-scalar algorithm. An eigenvalue or determinant alone is not a speedup.

The exact candidate and generic walks are deliberately an open pre-dispatch gate: a versioned amendment must freeze state/coefficient invariants, partition and update rules, increment derivation, canonicalization and baseline negation, distinguished points, collision recovery, success/termination, memory, parallelism, targets, and whole-attack accounting. Until then, the planted panel is functional/map holdout evidence only and cannot produce a DLP speedup or weakness label.

Controls and statistical confidence

Positive controls include the $(D=23)$ order-three norm-2 relation. The P-192 $(\sqrt{D}=2\pi-t)$ row is a deliberately partial control: its factorization is $(5 \cdot 11 \cdot 31 \cdot C)$, algebra must pass, and map/payoff must reject its unavailable (C)-degree support and scalar action on $(E(\mathbf{F}_p)[n])$. Ramified squares, inert primes, wrong orientations, unmatched large-prime columns, twist/UID substitutions, subgroup mutations, and map mutations are negative controls.

The scalar reference scan and segmented sieve must agree byte for byte on a small admitted box. Forward/reverse enumeration and one-worker/eight-worker sharding must recombine to identical roots. All-ones, shuffled, and perturbed Q32 weights may move ranks but not accepted relation IDs. A/A timing and operation controls must be consistent with parity.

The deterministic census is a finite population: report exact counts and digests, with no binomial interval and no extrapolation outside the boxes. Smoothness-model comparisons are descriptive. Timing claims use 31 paired public planted targets at interval widths 28, 32, and 36, identical targets, ABBA order, 10,000 target-cluster bootstrap resamples, and exact Clopper–Pearson success intervals. Width 36 COLD is primary; HOT and COLD are co-primary for the claim. A fresh clean rebuild on an independent environment must agree in direction.

Calibration, relation discovery, and solver outputs are selection data. Exactly one directed candidate and consumer are frozen before planted target scalars or timing-order material are derived. More than one holdout candidate requires a new preregistered familywise rule; choosing the fastest afterward is invalid.

Claim ladder and stopping rules

- **ABSTRACT_RELATION_FOUND**: exact principal row plus independent ideal/HNF replay.
- **DIRECTED_MAP_CHAIN_VERIFIED**: complete explicit maps, terminal isomorphism, subgroup and planted-point action.
- **RANKED_CANDIDATE**: Q32 order only; no speed claim.
- **SCOPED_ADVANTAGE**: exact consumer gate closed; HOT and COLD paired 95% upper endpoints below 1; 31/31 primary functional successes; controls and independent replication.
- **MATERIAL_WEAKNESS_CANDIDATE**: additionally conservative HOT and COLD ratios at most 0.80 with a verified action mechanism.
- **WEAK_CURVE_WITHIN_SCOPE**: full-order derivation gives at least 1.0 conservative security bit loss, ratio at most 0.5, interval lower bound above 1 bit, and independent replication.

A bounded planted solve alone never establishes full-order P-192 weakness. A timeout, OOM, missing map, solver cap, or partial shell is **INCOMPLETE**. A valid full null can say only **NO_WEAKNESS_FOUND_WITHIN_SCOPE** for these boxes, bases, bounds, solvers, and maps.

The family has eight frozen pending roles: preflight/controls, three sequential box shells, exact recombination and solver panel, map/cost replay, matched planted panel, and independent rebuild/replication. Canonical RUN IDs are allocated and rechecked only at dispatch; content-only names are not reservations. All are pending. A new Coordinator dispatch decision must bind final clean source and protocol commits before any run manifest exists.

Requirement-to-evidence matrix

- **Experiment, decision, and run family**: ledger records, experiment spec, and `run-family.yaml` are complete as protocol; every role is pending.
- **Exact P-192 tuple and identities**: `inputs.curve` is specified; execution replay is pending.
- **Discriminants, factorization, and maximal order**: `inputs.cm_order` contains the local identities; native certificate replay is pending.
- **Centered boxes and ordering**: `centered_norm_search` is specified and its raw counts pass the static checker.
- **Norm, smoothness, and LP rules**: `factor_bases_and_smoothness` is specified and unexecuted.
- **Encoding, orientation, and recombination**: `relation_encoding_and_recombination` is byte-exact; implementation is pending.
- **HOT/COLD and Q32**: `cost_contract` is specified; there are no calibration data.
- **LLL/BKZ/ILP**: `weighted_solver` fixes choices and budgets; no solver ran.
- **Maps, subgroup, and terminal certificates**: schemas are specified; there is no candidate.
- **Controls, statistics, stop and claim rules**: frozen but not exercised.
- **Archive outputs**: the protocol bundle and SHA256SUMS are present; all scientific outputs are pending.

Canonical-knowledge disposition

The current `knowledge/frontiers/ecdlp` material and the P-192 archive PR were checked as context. No canonical graph or knowledge finding changes here: there is no new measured relation, map, speedup, or obstruction. Promotion is expressly deferred until an independently replayed run supports a scoped claim.

Immediate gate

The next action is implementation admission, not a scientific search: add the segmented sieve, exact independent ideal verifier, partial/high-column kernels, directed map and cost records, optimizers, supervised runner, and independent replayer to the native repository; freeze the exact matched DLP consumer in a versioned amendment; pin and review the commits; then create a fresh dispatch decision. Until those gates, every result field remains empty.